



AJKD Atlas of Renal Pathology: Calcineurin Inhibitor Nephrotoxicity

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Clinical and Pathologic Features

Calcineurin inhibitor (CNI) nephrotoxicity may occur any time after initiation of therapy, and can affect transplant or native kidneys and present as acute or chronic reductions in kidney function. Elevated serum CNI trough levels may help make the diagnosis, although CNI nephrotoxicity is not necessarily dose dependent. CNI can cause vasoconstriction and increased serum creatinine (Scr) in the absence of morphologic findings, so CNI toxicity can be suggested as the cause of an abrupt Scr rise without morphologic changes. CNI nephrotoxicity-related acute injury is reversible with change in immunosuppression therapy; chronic injury is not reversible.

Light microscopy: Vasoconstriction-related CNI nephrotoxicity shows no morphologic changes. Isometric vacuolization of proximal tubular epithelium, and vascular injury with loss of smooth muscles, myocyte cytoplasmic vacuolization, and dropout from necrosis or apoptosis are characteristic of acute CNI nephrotoxicity. Ultimately, injured myocytes are replaced by focal nodular hyalinosis and eventual hyalinosis, which can have a distinctive adventitial localization and can extend to the media of the entire vascular wall of arterioles and arteries. CNI nephrotoxicity can also cause thrombotic microangiopathy (TMA) involving mostly arterioles and glomerular tufts. Chronic CNI toxicity shows a striped pattern interstitial fibrosis with proportional tubular atrophy.

Immunofluorescence microscopy: IgM and C3 staining in arterioles with hyalinosis and fibrinogen staining of thrombi in small arteries, arterioles, and glomeruli involved by thrombotic microangiopathy.

Electron microscopy: Isometric vacuoles in proximal tubular epithelium as dilated endoplasmic reticulum and large lysosomes. Hyaline accumulation within the

adventitia or media of arterioles and arteries. Glomerular endothelial swelling, expansion of lamina rara interna, and mesangiolysis are present in cases with TMA.

Etiology/Pathogenesis

CNI induces the release of vasoconstrictors, including endothelin, causing ischemic injury. The marked susceptibility to ischemia of the medullary rays gives rise to the striped pattern fibrosis, which also can occur due to ischemia of other causes. Afferent arteriolar vasoconstriction gives rise to functional CNI nephrotoxicity without morphologic change. CNI can also cause direct toxicity to vascular endothelium, vascular smooth muscle cells, and tubular epithelium. Injury may not be dose related.

Differential Diagnosis

Tubular injury from other etiologies, including osmotic tubular injury, ischemic acute tubular injury, and tubular injury associated with lipiduria in nephrotic syndrome can cause cytoplasmic vacuolization of proximal tubular epithelial cells.

TMA related to antibody-mediated rejection must be excluded, and usually displays significant microcirculation inflammation and peritubular capillary C4d staining and the patient has donor-specific antibodies. Other possible etiologies of TMA, including hemolytic uremic syndrome/thrombotic thrombocytopenic purpura, malignant hypertension, recurrent disease (if applicable), and antiphospholipid antibody syndrome also should be considered.

Hyalinosis of arterioles and arteries occurs in hypertension and diabetes-related injury. Hypertension-related hyalinosis is classically subendothelial and intimal; however, nodular medial hyalinosis does occur, particularly in more severe hypertension. Diabetic nephropathy has hyalinosis of both afferent and efferent arterioles. Transmural hyalinosis in advanced disease can make distinguishing different etiologies difficult.

Key Diagnostic Features

- CNI exposure
- Morphologic lesions may be absent (with functional CNI-induced vasoconstriction causing reduced kidney function)
- Nodular hyalinosis, particularly adventitial hyalinosis, extending to the media of arterioles and arteries

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- Striped interstitial fibrosis
- Thrombotic microangiopathy
- Isometric tubular vacuolization (less specific)

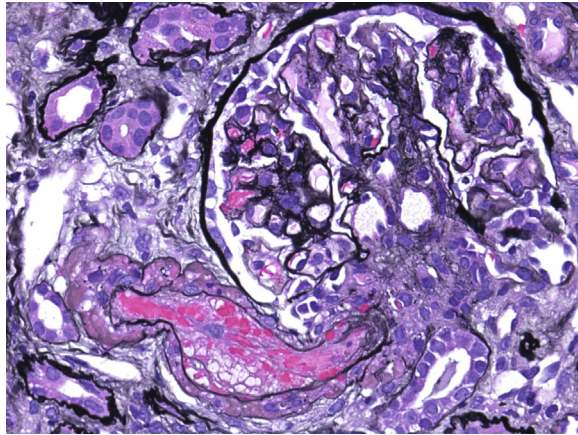


Figure 1. CNI nephrotoxicity with TMA involving an arteriole (Jones silver stain).

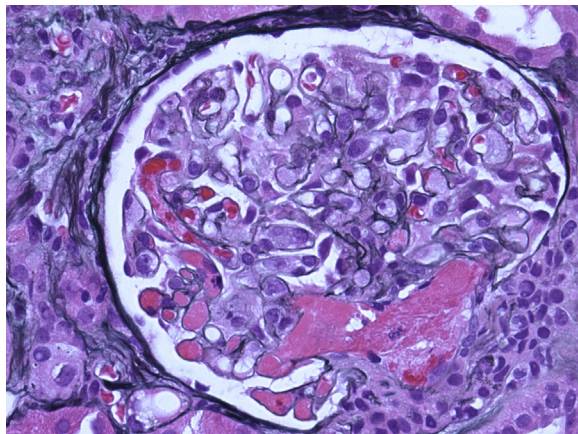


Figure 2. CNI nephrotoxicity with TMA involving the glomerular tuft and the vascular pole (Jones silver stain).

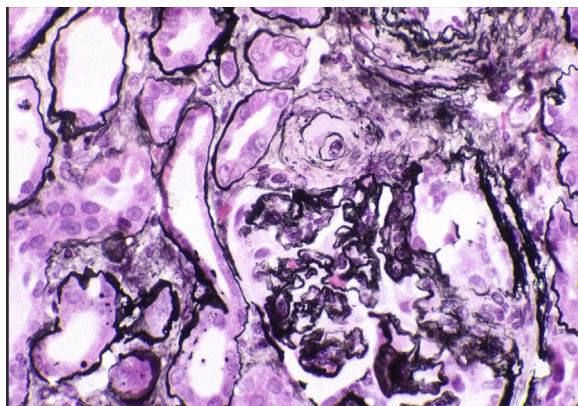


Figure 3. CNI nephrotoxicity with hyaline extending to the media of an arteriole and segmentally sclerosed glomerulus superimposed on ischemic change (Jones silver stain). Reproduced with permission from *AJKD* 36(1):e1-e2.

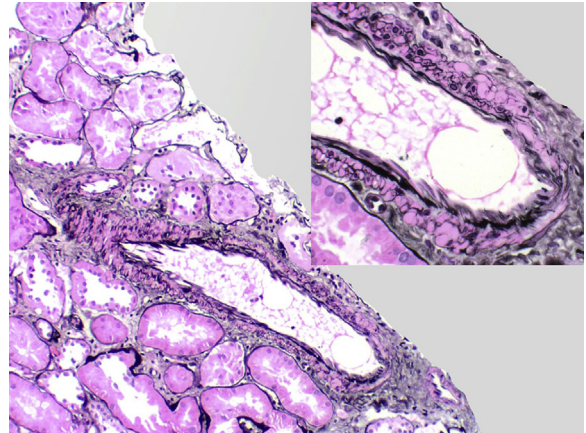


Figure 4. CNI nephrotoxicity with nodular hyalinosis extending to the media of an interlobular artery (Jones stain).

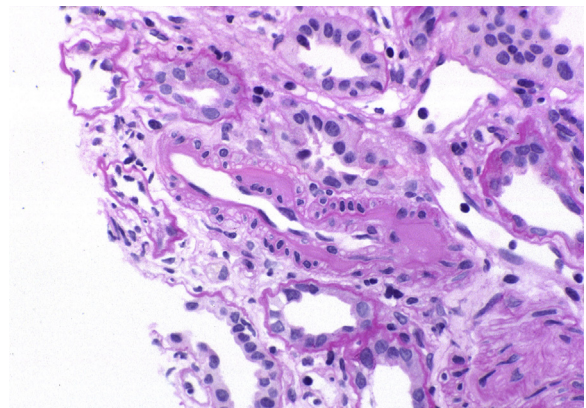


Figure 5. CNI nephrotoxicity with nodular hyalinosis extending to the media of an arteriole (periodic acid-Schiff stain). Reproduced with permission from *AJKD* 36(1):e1-e2.

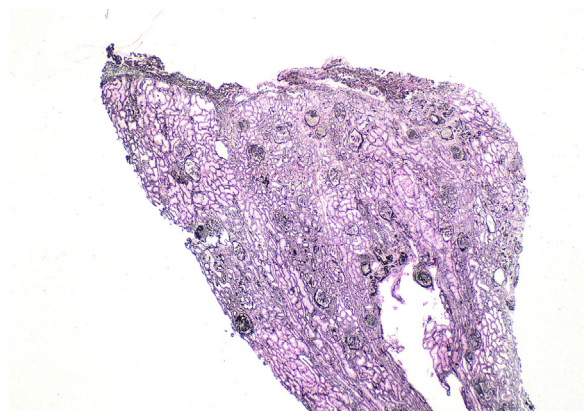


Figure 6. CNI nephrotoxicity with striped interstitial fibrosis (Jones silver stain). Reproduced with permission from *AJKD* 36(1):e1-e2.